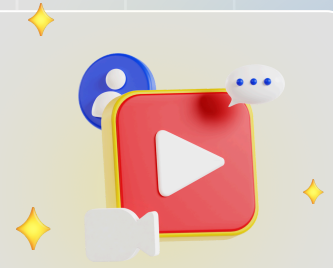


Case Study – AI Video Analytics

Real-Time Visual AI

AI-Powered Intelligent Monitoring at Scale



OVERVIEW

InfoObjects implemented a real-time video analytics solution within a multi-departmental organization. The engagement focused on designing and deploying a scalable AI pipeline capable of detecting and classifying objects from live video streams across various business functions. The pipeline enabled the client to extract operational insights, automate monitoring, and lay the groundwork for long-term analytics and model enhancements.



“InfoObjects’ experience with Generative AI has helped us implement a robust, AI-powered evaluation module that provides our instructors with practical, actionable insights. This innovative approach has enabled us to pinpoint specific areas for improvement, leading to measurable gains in classroom engagement and overall teaching effectiveness.”

—Vice President,
Corporate Training

CHALLENGE

Despite having extensive video surveillance infrastructure, the client faced multiple operational limitations:

- Lack of real-time situational awareness across facilities
- High reliance on manual monitoring and delayed response times
- No structured format for utilizing video data in analytics or reporting
- Difficulty detecting edge cases such as low-visibility or rare events

SOLUTION

The solution seamlessly captures live video feeds and intelligently analyzes each frame in real time to identify and categorize objects with high accuracy. It automatically logs key event data in a structured format, enabling teams to gain instant visibility into on-the-ground activity. By transforming raw video into actionable insights, the system empowers organizations to make faster, smarter decisions, enhance operational awareness, and proactively address issues as they arise.

APPROACH

InfoObjects implemented a YOLO-based object detection pipeline optimized for real-time performance and scalability. The pipeline included:

- **Annotation & Labeling:** Manual annotation using CVAT/Roboflow, or custom-built tools to generate high-quality training datasets tailored to the client's use cases.
- **Model Training & Optimization:** Training YOLO models with custom datasets, followed by performance tuning using NVIDIA GPUs and TensorRT for low-latency inference.
- **Real-Time Detection Integration:** Live RTSP video stream ingestion via FFmpeg and OpenCV with simultaneous multi-stream support.
- **Structured Logging & Recording:** Captured metadata including bounding boxes, class labels, confidence scores, and timestamps in structured formats (CSV/JSON), with MP4 recordings for audits and replays.

CONCLUSION

By implementing a real-time AI-powered detection pipeline, InfoObjects enabled the client to transform how video data is used across business operations. What was once unstructured and underutilized is now a continuous stream of actionable intelligence—fueling faster decisions, increasing safety, improving efficiency, and unlocking new levels of operational insight.

This project demonstrated the value of smart video analytics beyond traditional surveillance, showcasing its potential as a strategic asset across industries. With a scalable foundation in place, the client is now positioned to expand intelligent automation and analytics across even more functions in the future.

Free Generative AI Advisory Session

GenAI@InfoObjects.com

Contact us to schedule your session today.